

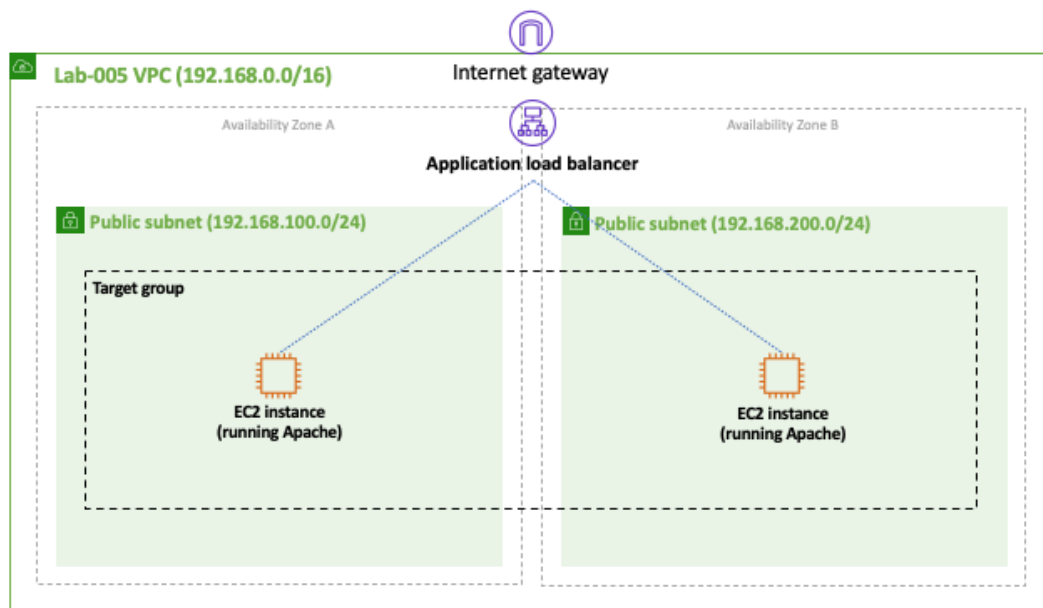
Lab-005

An HTTP Load Balancer

Goal

The goal of this lab is to illustrate how to use an [Application Load Balancer](#) to distribute requests over two web servers running on distinct Availability Zones. [Here](#) is a mind map of ELB (Elastic Load Balancing).

Architecture Diagram



Overview

AWS Elastic Load Balancing (ELB) service supports three types of load balancers:

- Application Load Balancer,
- Network Load Balancer, and
- Classic Load Balancer.

As the name suggests, the *Application Load Balancer* works at the application layer and understands protocols such as HTTP. This gives the load balancer the ability to support routing rules based on the URL or the host field in the HTTP header, for example.

Step 1 - Create VPC and Subnets

Create a new VPC with the *Name tag* lab-005 and the *IPv4 CIDR block* 192.168.0.0/16. Then create two public subnets in distinct availability zones of your VPC. Make sure to set the CIDR IPv4 block correctly to each subnet according to the ranges specified in the architecture diagram. You can use tags subnet-A and subnet-B for the subnets.

Create VPC

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances. You must specify an IPv4 address range for your VPC. Specify the IPv4 address range as a Classless Inter-Domain Routing (CIDR) block; for example, 10.0.0.0/16. You cannot specify an IPv4 CIDR block larger than /16. You can optionally associate an IPv6 CIDR block with the VPC.

Name tag ⓘ

IPv4 CIDR block* ⓘ

IPv6 CIDR block ☒ No IPv6 CIDR Block ⓘ
☐ Amazon provided IPv6 CIDR block

Tenancy ⓘ

* Required

Cancel

Create

Create subnet

Specify your subnet's IP address block in CIDR format; for example, 10.0.0.0/24. IPv4 block sizes must be between a /16 netmask and /28 netmask, and can be the same size as your VPC. An IPv6 CIDR block must be a /64 CIDR block.

Name tag ⓘ

VPC* ⓘ

Availability Zone ⓘ

VPC CIDRs	CIDR	Status	Status Reason
	192.168.0.0/16	associated	

IPv4 CIDR block* ⓘ

* Required

Cancel

Create

Create subnet

Specify your subnet's IP address block in CIDR format; for example, 10.0.0.0/24. IPv4 block sizes must be between a /16 netmask and /28 netmask, and can be the same size as your VPC. An IPv6 CIDR block must be a /64 CIDR block.

Name tag ⓘ

VPC* ⓘ

Availability Zone ⓘ different than subnet-A

VPC CIDRs	CIDR	Status	Status Reason
	192.168.0.0/16	associated	

IPv4 CIDR block* ⓘ

* Required

Cancel

Create

Create subnet

Actions ^

Filter by tags

☐	Name	State
☐	subnet-B	available
☒	subnet-A	available
☐	private	available
☐	public	available

Delete subnet

Create flow log

Modify auto-assign IP settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Share subnet

Add/Edit Tags

[Subnets](#) > Modify auto-assign IP settings

Modify auto-assign IP settings

Enable the auto-assign IP address setting to automatically request a public IPv4 or IPv6 address for an instance launched in this subnet. You can override the auto-assign IP settings for an instance at launch time.

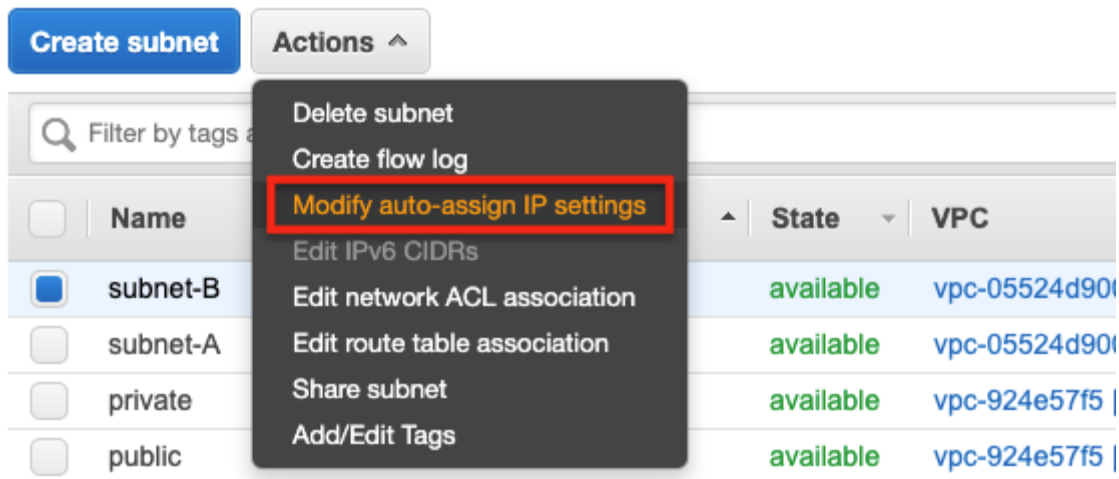
Subnet ID subnet-0d6308c2997abf63f

Auto-assign IPv4 ☒ Enable auto-assign public IPv4 address ⓘ

* Required

Cancel

Save



[Subnets](#) > Modify auto-assign IP settings

Modify auto-assign IP settings

Enable the auto-assign IP address setting to automatically request a public IPv4 or IPv6 address for an instance launched in this subnet. You can override the auto-assign IP settings for an instance at launch time.

Subnet ID subnet-0c4bd95216055242c

Auto-assign IPv4 ☒ Enable auto-assign public IPv4 address ⓘ

* Required

[Cancel](#)

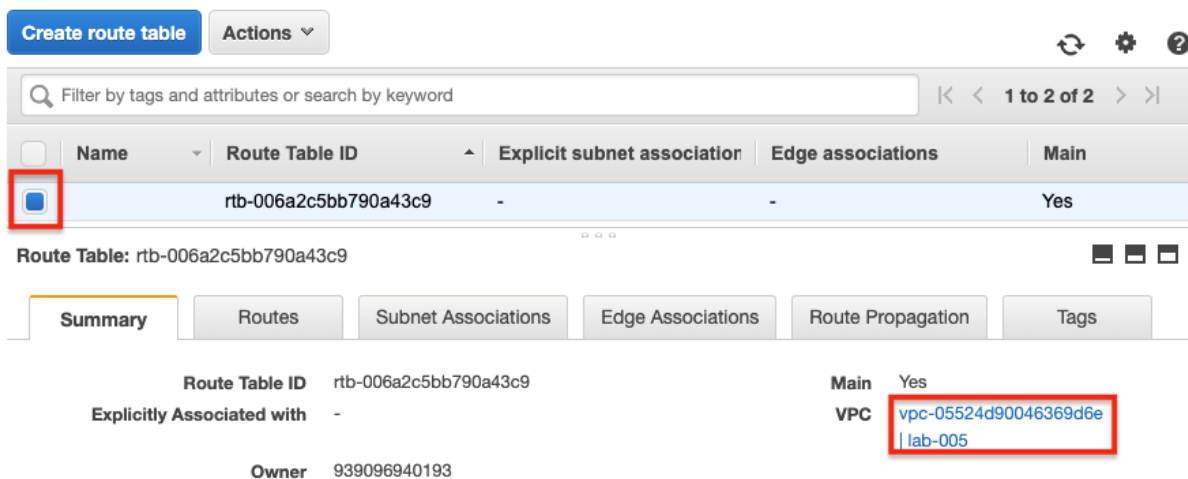
[Save](#)

Step 2 - Create an Internet Gateway

Create an [Internet gateway](#) with the *Name tag* lab-005 and attach it to lab-005 VPC. This procedure is described in detail in step 2 of [lab-002](#).

Step 3 - Modify the Main Route Table

Add a default route on your VPC router with a destination to the *Internet gateway* created in the previous step. Go to Subnet Associations and associate both subnets to the Main Route Table.



Create route table Actions

Filter by tags and attributes or search by keyword

Name	Route Table ID	Explicit subnet association	Edge associations	Main
	rtb-006a2c5bb790a43c9	-	-	Yes

Route Table: rtb-006a2c5bb790a43c9

Summary Routes Subnet Associations Edge Associations Route Propagation Tags

Edit routes

View All routes

Destination	Target	Status	P
192.168.0.0/16	local	active	N

Route Tables > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No

Add route

No results found.

* Required Cancel Save routes

Route Tables > Edit routes

Edit routes

Destination	Target	Status	Propagated
192.168.0.0/16	local	active	No
0.0.0.0/0	igw-		No

Add route

igw-09b59363a09b0b2f8 lab-005

* Required Cancel Save routes

Step 4 - Launch EC2 Instances

Launch an EC2 instance on each of the public subnets you created in the first step of this lab.

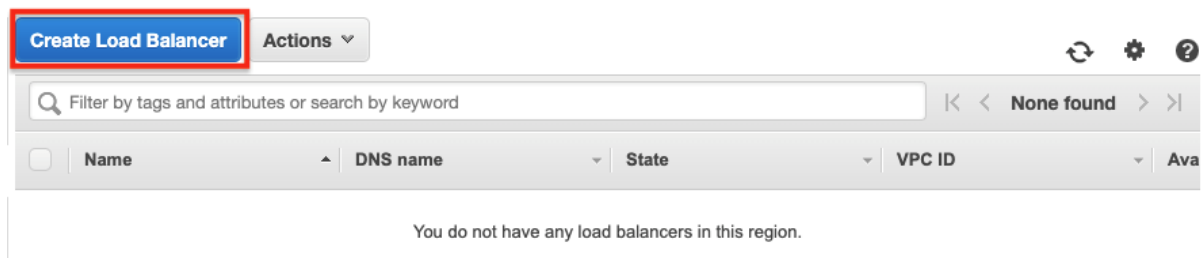
1. You can use *Amazon Linux 2 AMI*.
2. You can choose the *Free tier eligible* instance type.
3. Make sure to set the correct VPC and subnet for each instance; also, use the provided [user-data](#) to setup an HTTP server on your instances.
4. You can select the suggested storage for your instances.

5. You should add a *Name tag* to help identifying your instances.
6. Both instances should be in a *Security Group* with inbound rules that accept SSH and HTTP traffic.
7. You should secure your instances by creating a *lab-005* key pair.

At this point you should be able to access both instances using a browser on the instances public IP addresses. You should be able to see the hostname of each instance as the index page.

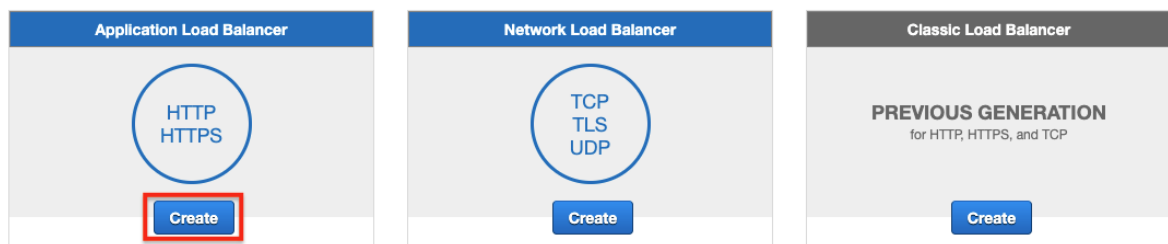
Step 5 - Create the Application Load Balancer

Follow the procedure described by the screenshots.



Select load balancer type

Elastic Load Balancing supports three types of load balancers: Application Load Balancers, Network Load Balancers (new), and Classic Load Balancers. Choose the load balancer type that meets your needs. [Learn more about which load balancer is right for you](#)



Step 1: Configure Load Balancer

Basic Configuration

To configure your load balancer, provide a name, select a scheme, specify one or more listeners, and select a network. The default configuration is an Internet-facing load balancer in the selected network with a listener that receives HTTP traffic on port 80.

Name ⓘ

Scheme ⓘ ☒ internet-facing
☐ internal

IP address type ⓘ

Listeners

A listener is a process that checks for connection requests, using the protocol and port that you configured.

Load Balancer Protocol	Load Balancer Port
HTTP	80
Add listener	

Availability Zones

Specify the Availability Zones to enable for your load balancer. The load balancer routes traffic to the targets in these Availability Zones only. You can specify only one subnet per Availability Zone. You must specify subnets from at least two Availability Zones to increase the availability of your load balancer.

VPC ⓘ

Availability Zones

☒ us-west-1a	subnet-0d6308c2997abf63f (subnet-A)	IPv4 address ⓘ	Assigned by AWS
☒ us-west-1c	subnet-0c4bd95216055242c (subnet-B)	IPv4 address ⓘ	Assigned by AWS

[Cancel](#) [Next: Configure Security Settings](#)

Step 3: Configure Security Groups

A security group is a set of firewall rules that control the traffic to your load balancer. On this page, you can add rules to allow specific traffic to reach your load balancer. First, decide whether to create a new security group or select an existing one.

Assign a security group: ☐ Create a new security group
☒ Select an existing security group

Filter

Security Group ID	Name	Description	Actions
☐ sg-0b906c2527a33aadd	default	default VPC security group	Copy to new
☒ sg-06c54e7b8b6047578	web-access	web-access	Copy to new

[Cancel](#) [Previous](#) [Next: Configure Routing](#)

Step 4: Configure Routing

Your load balancer routes requests to the targets in this target group using the protocol and port that you specify, and performs health checks on the targets using these health check settings. Note that each target group can be associated with only one load balancer.

Target group

Target group

Name

Target type ☒ Instance ☐ IP ☐ Lambda function

Protocol

Port

Health checks

Protocol

Path

Advanced health check settings

Cancel Previous **Next: Register Targets**

Instances

To register additional instances, select one or more running instances, specify a port, and then click Add. The default port is the port specified for the target group. If the instance is already registered on the specified port, you must specify a different port.

Add to registered on port

Add both instances to the target group!

Search Instances

☐	Instance	Name	State	Security groups	Zone	Subnet ID	Subnet CIDR
☒	i-0217be37f2f31558	instance A	running	web-access	us-west-1a	subnet-0d6308c2997abf63f	192.168.100.0/24
☒	i-0355937e2f9790de6	instance B	running	web-access	us-west-1c	subnet-0c4bd95216055242c	192.168.200.0/24

Step 5: Register Targets

Register targets with your target group. If you register a target in an enabled Availability Zone, the load balancer starts routing requests to the targets as soon as the registration process completes and the target passes the initial health checks.

Registered targets

To deregister instances, select one or more registered instances and then click Remove.

Remove

☐	Instance	Name	Port	State	Security groups	Zone
☐	i-0217be37f2f31558	instance A	80	running	web-access	us-west-1a
☐	i-0355937e2f9790de6	instance B	80	running	web-access	us-west-1c

Instances

To register additional instances, select one or more running instances, specify a port, and then click Add. The default port is the port specified for the target group. If the instance is already registered on the specified port, you must specify a different port.

Add to registered on port

Search Instances

☐	Instance	Name	State	Security groups	Zone	Subnet ID	Subnet CIDR
☒	i-0217be37f2f31558	instance A	running	web-access	us-west-1a	subnet-0d6308c2997abf63f	192.168.100.0/24
☒	i-0355937e2f9790de6	instance B	running	web-access	us-west-1c	subnet-0c4bd95216055242c	192.168.200.0/24

Cancel Previous **Next: Review**

Step 6: Review

Please review the load balancer details before continuing

▼ Load balancer

Name

lab-005

Scheme

internet-facing

Listeners

Port:80 - Protocol:HTTP

IP address type

ipv4

VPC

vpc-05524d90046369d6e (lab-005)

Subnets

subnet-0d6308c2997abf63f (subnet-A), subnet-0c4bd95216055242c (subnet-B)

Tags

Edit

▼ Security groups

Security groups

sg-06c54e7b8b6047578

Edit

▼ Routing

Target group

New target group

Target group name

lab-005

Port

80

Target type

instance

Protocol

HTTP

Health check protocol

HTTP

Path

/

Health check port

traffic port

Healthy threshold

5

Unhealthy threshold

2

Timeout

5

Interval

30

Success codes

200

Edit

▼ Targets

Instances

i-0217be37f2ff31558 (instance A):80, i-0355937e2f9790de6 (instance B):80

Edit

Cancel

Previous

Create

Load Balancer Creation Status

✓

Successfully created load balancer

Load balancer [lab-005](#) was successfully created.

Note: It might take a few minutes for your load balancer to be fully set up and ready to route traffic, and for the targets to complete the registration process and pass the initial health checks.

Suggested next steps

- Discover other services that you can integrate with your load balancer. Visit the **Integrated services** tab within [lab-005](#)
- Consider using AWS Global Accelerator to further improve the availability and performance of your applications. [AWS Global Accelerator console](#)

Close

Test and Validation

Copy the DNS name associated with the load balancer and, using a browser, try to access it. You should be able to see alternate responses (as you refresh the page) of the two instances webpages which proves that the load balancer is working properly by redirecting the request evenly to the two instances.

Create Load Balancer

Actions

Filter by tags and attributes or search by keyword

	Name	DNS name	State	VPC ID
☒	lab-005	lab-005-790551193.us-west-...	active	vpc-0552...

Load balancer: lab-005

Description

Listeners

Monitoring

Integrated services

Tags

Basic Configuration

Name

lab-005

ARN

arn:aws:elasticloadbalancing:us-west-1:939096940193:loadbalancer/app/lab

DNS name

lab-005-790551193.us-west-1.elb.amazonaws.com 

(A Record)

State

active

Another test you should do is to stop one of the instances and verify that the load balancer will not send requests to the stopped instance anymore.